Symposium on the personalized medicine of COPD

Philipps-Universität Marburg

Systematic update of a health economic Markov model for COPD using recent German cohort data

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Background and aims

Long-term consequences of interventions among COPD patients are typically simulated using Markov models. Reasons for using a Markov model:

- Wide range of scenarios can be analyzed.
- Assumptions are made explicit, their impact on the results can be investigated.

General objective:

To analyze the impact of a personalized COPD intervention on:

- Resulting direct and indirect costs,
- Health-related quality of life.

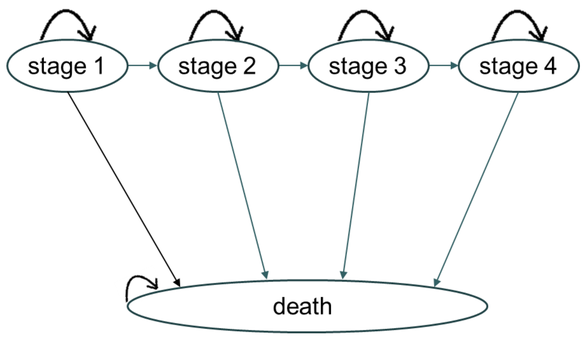
→ Long term projection of cost-effectiveness of personalized treatment in COPD (Subproject 5, work package 3).

Specific aim:

Update of an existing Markov model with respect to model structure, model parameters and software implementation

Methods

- Transfer of the spreadsheet-based model to R (open source, easier to maintain and replicate).
- Simplification of the model structure (reduced complexity of parameter estimations).
- Validation of change in software and model structure.
- Update of model parameters using own and external study data.
- Data from COSYCONET used to derive disease progression rates, exacerbation risk, utilities, costs, and mortality parameters.



Acute exacerbations are modeled as events within disease stages

Model simplifications: Omitted lung volume reduction and lung transplantation as states, as well as mild exacerbations as events. No distinction between smokers and non-smokers.

Results I: Mortality

Step 1: Estimation of stage-specific hazard ratios compared to normal population.

Method: Network meta-analysis of 12 studies

Stage	Network meta-analysis: Pooled HR reference normal
GOLD I	1.11 (0.96, 1.27)
GOLD II	1.60 (1.38, 1.88)
GOLD III	2.39 (1.95, 2.94)
GOLD IV	3.66 (2.90, 4.64)

Step 2: Age- and sex-specific mortality rates are calculated using German mortality data.

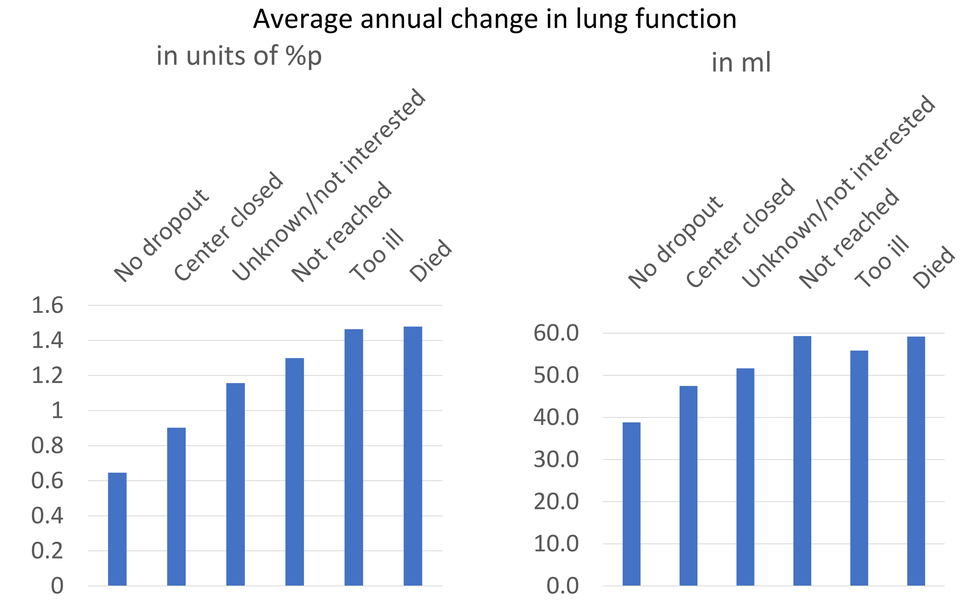
Example: Resulting 3-month mortality(%) for COPD patients aged 65-69 years:

	GOLD 1	GOLD 2	GOLD 3	GOLD 4
Men	0.51	0.73	1.09	1.66
Women	0.26	0.38	0.57	0.87

Results II: Disease progression

Step 1: Estimation of lung function decline from COSYCONET data.

Method: Linear mixed models with data of up to 6 years follow-up taking drop-outs into account



Estimated annual change for GOLD 1, 2, 3 stages respectively: -1.00, -1.03, -0.77 FEV1 %pred.
→ will be transformed into progression rates for each GOLD stage (Step 2, to be done)

Summary and outlook

Longitudinal COPD cohorts like COSYCONET in combination with other data sources can be used to update the parameters of a COPD Markov model, such as progression rates, mortality rates, utilities and costs. Together with clinical trial data on effectiveness of personalized treatment, this will allow long term projections of cost-effectiveness.

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